

PART 1: "TrackSwap" — The Airbnb for Racing Circuits

1.1 Product Vision

TrackSwap is a two-sided marketplace connecting track owners (private circuits, karting tracks, airfields with tarmac, industrial lots suitable for autocross) with drivers seeking track time. Many private circuits sit empty 60–70% of available days. Many drivers can't find affordable or available track time because the well-known circuits are booked solid on popular dates. TrackSwap matches supply with demand and handles all the friction: insurance, booking, safety briefings, noise compliance, and payment.

Core Insight: Track day demand has grown 40%+ since 2020 in Europe and the US. Yet track supply is constrained and fragmented. In most European countries, there are 5–15 'official' circuits and hundreds of unused venues that could host driving events—private karting tracks that close on weekdays, agricultural airfields with perfect tarmac used 10 days a year, industrial estates with empty parking lots on weekends. This hidden inventory represents a massive untapped supply-side opportunity.

USP: The only platform that discovers, onboards, and makes bookable non-traditional driving venues with integrated per-session insurance, digital safety compliance, noise monitoring, and dynamic pricing. TrackSwap is not a ticket seller—it's an infrastructure layer that turns underutilized asphalt into bookable driving experiences.

Target Audience: Supply: Private circuit owners, karting facility operators, airfield owners, industrial estate managers. Demand: Track day enthusiasts, driving experience companies, racing schools, corporate event organizers, automotive press, film production companies.

1.2 Competitive Landscape

Market Overview

The track day booking market is fragmented across regional providers. Key players include TrackDays.co.uk (UK's largest aggregator, lists events from multiple organizers), BookaTrack/Want2Race (UK, transitioned from premium track days to racing experiences), Trackday-Hub.com (European calendar aggregator covering Silverstone, Spa, Nürburgring), EuropaTrackdays.com (pan-European calendar), and individual Track Day Organizers (TDOs) like RMA Track Days, Goldtrack, Opentrack, and Lotus on Track. In the US, market is even more fragmented with NASA (National Auto Sport Association), SCCA, and regional track day organizers. No platform operates as a true marketplace with supply-side onboarding.

Direct Competitor Matrix

Feature	TrackDays.co.uk	Trackday-Hub	BookaTrack	RMA/Goldtrack	NASA/SCCA	TrackSwap
Model	Aggregator	Calendar	TDO (organizer)	TDO (premium)	Membership org	Marketplace
Venue Onboarding	✗ Lists TDOs only	✗ Calendar only	✗ Own events	✗ Own events	✗ Own events	✓ Active supply
Non-Traditional Venues	✗ No	✗ No	✗ No	✗ No	✗ No	✓ Core USP
Integrated Insurance	✗ Third-party	✗ None	○ Basic	○ Basic	✗ Separate	✓ Per-session
Noise Monitoring	✗ No	✗ No	○ Manual dB	○ Manual dB	✗ No	✓ Real-time API
Dynamic Pricing	✗ Fixed	✗ N/A	✗ Fixed	✗ Fixed	✗ Fixed	✓ Demand-based
Digital Safety Brief	✗ In-person	✗ N/A	○ PDF	○ In-person	○ In-person	✓ In-app + video
Coverage	UK focus	Europe	UK	UK/EU	US	UK+EU launch, global
Revenue Model	Commission	Ads/listings	Ticket sales	Ticket sales	Membership	12–18% take rate

SWOT Analysis

STRENGTHS	WEAKNESSES
First-mover in venue marketplace model; unlocks hidden supply (airfields, kart tracks, industrial lots); integrated insurance removes #1 friction point for venue owners; network effects (more venues = more drivers = more venues); digital compliance stack creates switching costs; applicable to cars, motorcycles, karting, autocross, drifting, driving schools	Cold start: need venues AND drivers simultaneously; venue quality is unproven and variable; insurance partnerships complex to negotiate initially; noise regulations vary by country/municipality; seasonal demand in Northern Europe; liability concerns for non-traditional venues; no brand recognition at launch
OPPORTUNITIES	THREATS
Track day participation growing 40%+ post-COVID; thousands of underutilized venues across Europe; driving experience companies need flexible venue access; corporate events increasingly seeking 'experiential' venues; film production needs test locations; EV track days emerging (no noise limits!); adjacent: motorcycle track days, cycling velodromes, drone racing	TrackDays.co.uk could add marketplace features; established TDOs have loyal customer bases; venue owners may resist platform intermediation; insurance providers may be cautious with non-traditional venues; regulatory complexity across jurisdictions; Airbnb/Hipcamp could theoretically enter motorsport

1.3 MVP Scope

MVP Goal

Validate that (a) venue owners will list on the platform and accept bookings, and (b) drivers will book through TrackSwap rather than directly with TDOs. Target: 30 venues listed, 500 bookings in first 6 months. Focus: UK + Spain (strong track day culture, year-round weather in Spain).

MVP Feature Set

Priority	Feature	Details
P0 — Core	Venue Listing & Onboarding	Venue owner creates profile: photos, surface type, dimensions, facilities, noise limits, location, availability calendar. Onboarding wizard with surface quality self-assessment.
P0 — Core	Search & Discovery	Driver searches by location, date, venue type (circuit/karting/airfield/lot), price range, noise limit. Map view with venue pins. Calendar view for availability.
P0 — Core	Booking & Payment	Stripe-powered booking. Venue owner sets base price per session. Platform takes 15% commission. Instant booking or request-to-book. Cancellation policies.
P0 — Core	Digital Safety Briefing	Mandatory pre-event safety video/checklist per venue. Driver must complete before arrival. Reduces venue liability. Digital waiver with e-signature.
P1 — Important	Insurance Integration	Per-session track day insurance: on-track damage cover (vehicle), third-party liability. One-click purchase at checkout. Partnership with specialist motorsport insurer.
P1 — Important	Venue Rating & Reviews	Drivers rate: surface quality, facilities, safety, value. Venue quality score. Photos from attendees. Builds trust for non-traditional venues.
P1 — Important	Noise Monitoring Dashboard	IoT-connected dB meter at venue boundary (optional hardware). Real-time noise level visible to venue owner and drivers. Auto-alert when approaching limit.
P1 — Important	Organizer Mode	For driving experience companies and racing schools: bulk booking, recurring schedules, branding on listing, attendee management.
P2 — Nice-to-Have	Dynamic Pricing	AI-suggested pricing based on day-of-week, weather forecast, historical demand, nearby events. Venue owner can accept or override.
P2 — Nice-to-Have	Group Booking	Split costs among friends. Minimum group size discounts. Private session option.

Post-MVP	Venue Certification Program	TrackSwap-certified venues get priority listing, insurance discounts, and quality guarantee badge.
Post-MVP	Corporate Events Module	Packages for team-building: catering, photographer, instructor add-ons. One-click corporate booking.

MVP Timeline & Budget

Phase	Duration	Deliverables
Phase 1: Platform Build	Weeks 1–8	Next.js web app (responsive), venue onboarding flow, search/map, Stripe booking, safety briefing system, admin dashboard.
Phase 2: Supply Seeding	Weeks 4–12	Recruit 30 venues: 10 UK circuits/kart tracks, 10 Spain circuits, 10 non-traditional (airfields, lots). Onboarding support. Photography.
Phase 3: Insurance Partner	Weeks 6–12	Partner with motorsport insurer (e.g., Reis Motorsport, Adrian Flux). API integration for instant quotes at checkout.
Phase 4: Beta Launch	Weeks 10–14	Invite 500 track day enthusiasts from forums (PistonHeads, SEAT Cupra forums, Spanish car clubs). Test booking flow.
Phase 5: Public Launch	Weeks 14–18	App Store + web launch. PR targeting automotive media. Referral program: invite a venue, get £50 credit.

Role	Count	Monthly Cost
Full-Stack Engineers (Next.js/Node)	2	€8,000–12,000 each
Product Designer	1	€5,000–8,000
Business Development (venue onboarding)	2 (per region)	€3,000–5,000 each
Founder / PM	1	Equity-only
Insurance Integration / Legal	—	€5,000–10,000 (one-time)
Cloud Infrastructure	—	€300–800/mo
Monthly Burn	—	€30,000–50,000
MVP Total (4.5 months)	—	€135,000–225,000

1.4 Technical Architecture

Technology Stack

Layer	Technology
Frontend	Next.js 14+ (SSR for SEO, critical for venue discovery). Tailwind CSS. MapLibre GL for venue map. Responsive-first (mobile booking is primary use case).
Backend	Node.js with tRPC or REST API. PostgreSQL with PostGIS for geospatial venue search. Redis for caching availability calendars.
Booking Engine	Custom state machine: Available → Reserved (15 min hold) → Paid → Confirmed. Stripe Connect for split payments (platform fee + venue payout). iCal export for venue owners.
Insurance API	REST integration with motorsport insurer. Quote engine: vehicle type + session duration + venue risk profile → instant premium. Policy issued at checkout.
Safety Compliance	Video-based safety briefing per venue (Cloudflare Stream or Mux). Completion tracking. Digital waiver via DocuSign API or custom e-signature. GDPR-compliant data handling.
Noise Monitoring	Optional IoT: Raspberry Pi + calibrated dB meter (SPL sensor) at venue boundary. MQTT to cloud. Real-time dashboard via WebSocket. SMS alerts to venue owner at threshold.
Search & Discovery	PostGIS radius search with venue type/noise/price filters. Elasticsearch for full-text search. Algolia for instant search UX.
Analytics	Mixpanel for product analytics. Custom dashboard for marketplace health metrics (supply/demand ratio per region, booking conversion, venue utilization rate).
Infrastructure	Vercel for frontend, Railway/Render or AWS for backend. Docker containers. GitHub Actions CI/CD. Terraform for IaC.

Core Data Model

Venues: id, owner_id, name, type (circuit/kart/airfield/lot), location (PostGIS), surface_type, dimensions, max_vehicles, noise_limit_db, facilities[], photos[], safety_video_url, insurance_risk_tier, rating, booking_count, status

Availability: id, venue_id, date, time_slots[], price_per_slot, max_capacity, booked_count, dynamic_price_override, weather_forecast

Bookings: id, venue_id, driver_id, slot_id, vehicle_type, insurance_policy_id, safety_brief_completed, waiver_signed, payment_id, status, total_price, platform_fee

Revenue Model

Stream	Rate	Details
Platform Commission	12–18% per booking	Standard marketplace take rate. Lower for high-volume venues.
Insurance Referral	10–15% of premium	Revenue share with insurer per policy sold through platform.
Premium Venue Listing	€49–99/mo	Priority placement, analytics dashboard, branded profile.
Corporate Events Module	€199–499/event	Custom branding, attendee management, catering coordination.
Noise Monitoring Hardware	€149 one-time + €9.99/mo	IoT dB meter rental/purchase with cloud dashboard.